Technical Document #869: Retrieval Augmented Generation - Comprehensive Overvi

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Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Query decomposition breaks complex queries into simpler sub-queries.
- Hybrid search combines keyword-based and semantic search.